

STAT 200: Introductory Statistics

Winter 2026

Instructor Information

Name: Torin Quinlivan

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Office Hours: Before Class or by Appointment

Class Information

Dates: January 5 – March 15

Time: Period 6: 2:40 – 3:50 p.m.

Classroom: Science + Math Center - E117

Course Description

The purpose of this course is to introduce you to analyzing data and performing statistical analyses in your own research. Briefly, this five-step process is:

1. framing the research,
2. collecting the data,
3. analyzing the data,
4. interpreting the results, and
5. presenting the results.

To support these steps, we will also cover some basics of probability math and probability distributions. The probability topics allow one to better understand the inherent randomness in life (and in our research).

Course Objectives

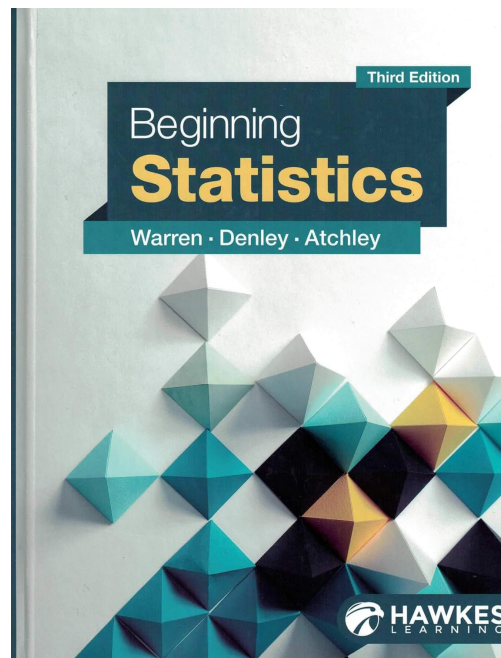
By the end of this course, you should:

- understand why the differences between the sample and the population are important;
- compare and contrast several data-collection techniques in terms of bias, variance, and cost;
- summarize the sample using appropriate graphical and numerical methods;
- analyze statistical writing for appropriateness;
- apply proper statistical methods to research questions; and
- evaluate several statistical methods, select, and perform the most appropriate one (or ones).

Textbook, Calculators, & Software

Textbook: Carolyn Warren, Kimberly Denley, and Emily Atchley. *Beginning Statistics*. Third ed. Mount Pleasant, SC: Hawkes Learning System, 2020. <https://www.hawkeslearning.com/Products/Math/BEG3/BeginningStatistics3e.html>

You will *not* need the homework software access, there is a bundle of software and eBook access without a physical copy or a bundle with a physical copy available, but as we will be using WebWork for our homework, you can simply find a used copy for sale to read.



Optional Text: Dr. Forsberg wrote a text about statistics and specifically R, if you'd like another option, and to learn more about application:

Ole J. Forsberg. *R for Starters*, v55721. Galesburg, IL, 2021. (<https://rfs.kvasaheim.com/>)

Calculator: You will need a calculator to do the computations that will arise throughout the course. No specific calculator is required; I recommend the TI-30XS MultiView (<https://a.co/d/0QHmuee>). It's cheap and can do anything you need to in this course. You don't need a graphing calculator for this course, but if you have a TI-84 already, it's more than capable.

Software: In this course, we will be using R for our statistical software. It's completely free, widely-used, easy to install, and is very flexible with many packages available. R can be downloaded at <https://cran.r-project.org/>, and I recommend installing and working in RStudio, found at <https://posit.co/download/rstudio-desktop/>. I'll go over installing and using R in our first Thursday class, so if you have any issues, feel free to bring them to me.

Class Attendance and Participation

It is essential to your success in this course that you attend and participate in each lecture. I will not take attendance directly, however there is an element of your grade that will include in-class elements. In my experience as a student, a teaching assistant and as a lecturer, the number one most common problem students have in courses is falling behind in the course and not being able to catch back up. If you are forced to miss a day, it is your responsibility to catch yourself back up, whether through getting notes from a classmate or coming to me to go over the material.

In class, I ask that you respect my time, your time and your classmate's time. In class, please avoid anything that would be too distracting your classmate: no cell phones, music, loud foods, etc. I personally recommend taking notes with pen and paper, but if you prefer using a laptop/tablet, please stay focused on the lecture and don't go surfing the web. (Do kids still say that?)

I aim to create an open and welcoming environment for learning, so if you have questions or don't understand something I've said, please speak up. If you are confused about something, it's almost certain that you're not the only one, and I'm happy to clarify. If you notice a mistake in my math, let me know, I'm only human and I promise I've made more stupid math mistakes than you have! And if you're curious or want to know more about a subject, speak up, I got my PhD in Mathematics & Statistics because I love this stuff!

Email Policy

If you have any questions outside of class, you can stop by my office, but you can also email me. I will attempt to respond to all emails within 24 hours. If you send an email late at night, I may not respond until the next day, and if you email on the weekends, I may not respond until the following Monday. If I fail to respond after 24 hours during the week, please resend the email, as I may have just not noticed it. I recommend that you email me using your Knox College email account, and keep note that there may be topics I may not be able to discuss over email or feel comfortable discussing over email, in which case I will make a time for us to meet in person to speak.

Academic Integrity & Honor Code

Remember that all work you submit must be in accordance with the Knox College Honor Code (<https://www.knox.edu/offices/academic-affairs/honor-code-and-procedures/>). Should I find evidence of a violation of academic integrity, I will sanction you.

Note two passages from the Honor Code in particular:

Students are expected to leave books and notes elsewhere during an examination and to take examinations in public places such as classrooms or open lounge areas nearby in the buildings. Washrooms, storage areas, and maintenance areas in the basement and elsewhere are not public places. Neither are carrels or other closed areas . . .

Faculty may require that students not bring books, notes or electronic devices into the building in which an examination is administered. Alternatively, they may permit students to bring these resources into the building but may require that these books, notes and electronic devices be left in a central area or the classroom throughout the duration of the examination or other academic obligation.

In accordance with these policies, you should not have a cell phone on you when taking exams or quizzes. I will not allow computers or other electronic devices, notes, textbooks, or any other help on exams.

Homework can be done cooperatively, but any submissions must be in your own words.

AIs such as Chat-GPT are expressly forbidden in solving the problem, both because they will not help you learn (the only way to learn mathematics is through doing mathematics), and because it doesn't even answer correctly most of the time, so don't waste your time. I am open to you using it as a learning aid, and would even be interested in hearing *how* you all use it in your studying, but you need to be able to solve the problems on the quizzes and the exams, so do not rely on it!

Office of Disability Support Services

The Office of Disability Support Services website can be found at <https://www.knox.edu/offices/disability-support-services>.

Knox College abides by Section 504 of the Rehabilitation Act of 1973 (<https://www.epa.gov/external-civil-rights/section-504-rehabilitation-act-1973>), which stipulates:

No otherwise qualified individual with a disability in the United States, as defined in section 7(20) shall, solely by reason of her or his disability, be excluded from the participation in, be denied the benefits of, or be subjected to discrimination under any program or activity receiving Federal financial assistance or under any program or activity conducted by any Executive agency or by the United States Postal Service.

Disabilities covered by law include, but are not limited to, learning disabilities, psychological disabilities, health impairments, hearing, and sight or mobility impairments.

If you have a disability that may have some impact on your work in class and for which you may require accommodations, please contact Stephanie Grimes in the Office of Disability Support Services (office: SMC E-115; email: sgrimes@knox.edu) so that such accommodations may be arranged.

Note that the accommodations are not retroactive; I do not make adjustments until ODSS contacts me to let me know what I can do. Also note that accommodations must be renewed each term.

Tutoring

The Center for Teaching and Learning (CTL) offers many resources for students in need of help. Asking for help is no reason to be embarrassed; seeking help is a big part of the learning process, and a sign that you are engaging with the material and your coursework. You are not alone at Knox College and you should utilize all the tools at your disposal to succeed.

Writing Tutors: If you want help on a paper, no matter the course, you can find writing tutors in the Writer's Workshop, which is at the Center for Teaching and Learning (466 S. West St., across from Post Hall), Monday through Friday, from 10:30 to 4:00pm, and again in Red Room on Tuesday, Wednesday, and Thursday nights from 7:00 to 9:00pm. You can also schedule appointments to meet with a writing tutor either online or in person (should you both so desire) by visiting the

Center for Teaching and Learning website.

Subject Tutors: The Red Room study tables are great resources for students wanting help with virtually any subject— including STAT 200. There are two Red Room study areas: “Red Room” and “Red Room SMC.” Red Room SMC is intended to support most classes taught in SMC and is located in the Learning Commons in SMC (under the whale), while Red Room is for most other courses and is located in the Red Room on the second floor of Seymour Library. Both study areas are free and open on a walk-in basis from 7:00pm to 9:00pm, on Tuesday, Wednesday, and Thursday nights. Specifically, this class, STAT 200, is tutored in the Red Room in SMC.

You can also get help from a tutor attached to our class by scheduling an appointment to meet them one-on-one. The attached CTL tutors:

- TBD (tbd@knox.edu)
- TBD (tbd@knox.edu)
- TBD (tbd@knox.edu)

They will stop by one day early in the semester to introduce themselves. Please remember that our tutors are students themselves, and as such please be considerate of them and respect their time.

TimelyCare

If you're unfamiliar with mental health support, taking care of your mental well-being can be stressful, or even scary. Knox is partnered with TimelyCare to provide a virtual health care solution that reflects Knox's values and student population. It is a student-centered platform that offers on-demand access to mental healthcare and a diverse, culturally competent provider network. TimelyCare complements Knox's on-campus services with 24/7 on-demand virtual mental health care, health coaching, peer to peer support and a plethora of coping strategies ranging from homesickness to depression to study skills. There are 12 free sessions available from September to August annually and can be accessed anywhere on or off campus (even internationally!). To access TimelyCare, go to [TimelyCare.com](https://www.timelycare.com) or use the QR code below, register with your Knox email and follow prompts! If you have any questions, please reach out to Knox Health and Wellness at counseling@knox.edu or (309) 341-7492. If there is an emergency, please call 911 or contact Campus Safety at (309) 341-7979.



Title IX

Knox College values a culture of respect. If you have encountered any form of discrimination or harassment, you are encouraged to report it at <https://www.knox.edu/about-knox/our-values/culture-of-respect/report-it>.

As a faculty member, I am required to disclose to the Director of Title IX & Civil Rights Compliance any information I receive about discrimination, harassment, sexual misconduct and other forms of interpersonal violence. This includes information shared in class assignments, discussions, emails, and/or shared conversations outside of class. If a disclosure is received, she will reach out to you. You are not required to respond to this outreach.

If you wish to speak with a confidential resource (one not required to report to Title IX), please contact:

- Knox College Health Services (<https://www.knox.edu/offices/health-and-counseling-services/health-services>)
- Knox College Counseling Services (<https://www.knox.edu/offices/health-and-counseling-services/counseling-services>)
- Knox College Director of Spiritual Life (<https://www.knox.edu/offices/spiritual-life/meet-the-staff>)
- TimelyCare (<https://app.timelycare.com/auth/login>)

Pregnancy and/or Pregnancy Related Conditions

Faculty members are required to inform a student who is pregnant or experiencing a pregnancy related condition (including termination) that the Director of Title IX & Civil Rights Compliance is available to provide support by helping you to understand your rights/options and provide supportive measures. You can contact the Director of Title IX & Civil Rights Compliance at cultureofrespect@knox.edu.

Grading Information

Exams

About halfway through the term, around the end of Module 3, we will have a midterm exam. The exact date of the exam will be decided later and announced in class, with ample time to prepare. It will count for **100 points**.

At the end of the course, we will have a cumulative final exam during the time scheduled by the registrar's office. It will count for **150 points**.

Exams will be given in the classroom that we do lectures in. I have spoken with Dean Crawford, and I will be expecting everyone to take the exam in the room, unless you have an accomodation and prior approval. A calculator is all you can use on the exam; no computers, no phones, no notes, not talking to other classmates.

WebWork Assignments

These assignments are located in the WebWork part of the course. These are due at 11:59pm on the due dates assigned and announced in class. I transfer the assignment grades to my gradebook after that. Late assignments are worth 50% and can be completed for one week after the due date. I will drop the lowest few and average the rest of the assignment grades for a total of **100 points**.

Statistics Labs

On Thursdays that are not exam days, we will have a statistics lab assignment or a quiz, something to evaluate your learning so far. These will focus on application and often utilize R. There will be eight of them, and I will drop the lowest one for a total of **100 points**.

In-Class Activities

During most lectures, I will try to assign small assignments, likely a practice problem done together or a small "quiz". You will need to complete about 20–22 of them to get full credit, for a total of **50 points** overall. If you attend fewer, you will receive the proportion out of the max required. Note that you should not *need* to attend all of the lectures to get max points here, but I urge you in the strongest possible way to please *please* attend all lectures.

Grading Breakdown

The grades will break down as follows:

Component	Points
Midterm Exam	100
Final Exam	150
WebWork	100
Labs/Quizzes	100
In-Class	50
Total	500

Grade Scale

Final grades will be assigned according to the following point totals:

Grade	Points
A	465
A-	450
B+	435
B	415
B-	400
C+	385
C	365
C-	350
D+	335
D	315
D-	300